ECO101 - Introduction to Microeconomics

Department of Economics and Social Science (ESS), Brac University

Recap

So far we discussed

- PFF and opportunity cost
- Absolute Advantage
- Comparative Advantage
- Gains from Trade

Now we will be looking into how markets work

  - How prices are determined?

Demand, Supply & Market Equilibrium
Lecture 2

Reading

- Economics by Michael Parkin, 10th edition
    - Chapter: 3
- Principles of Economics by Mankiw, 6th Edition
    - Chapter: 4

Introduction

- Understand how a market economy works - theory of demand and supply
- Demand and supply are the forces that make market economies work
    - They determine the quantity of each good produced and the price at which it is sold
    - If you want to know how any event or policy will affect the economy, you must think first about how it will affect supply and demand
- Demand and supply considers how buyers and sellers behave and how they interact with one another
- Demand and supply determine prices in a market economy and how prices, in turn, allocate the scarce resources in the economy

Introduction

- The terms supply and demand refer to the behavior of people as they interact with one another in competitive markets
- Competitive market refers to a market that has many buyers and many sellers, where no single buyer or seller can influence the price and must accept the price the market determines
  - **They are price takers**
- We are going to study how people respond to prices and the forces that determine prices

Demand

Demand

- The <u>quantity demanded</u> of a good or service is the amount of the goods that the buyers are willing and able to purchase in a given period at a particular price
- If a person “demands” something, they
  - Want it,
  - Can afford it, and
  - Plan to buy it

Law of Demand

- ➢ The Law of Demand states that, other things remaining the same,
  - The higher the price of a good, the smaller is the quantity demanded
  - The lower the price of a good, the greater is the quantity demanded

Why?

- ➢ For two reasons:
  - Substitution effect
  - Income effect

Substitution Effect & Income Effect

- ➢ Substitution effect
  - The change in the relative price of a good

- When the price of a good rises, other things remaining the same, it’s relative price rises

  ➔ opportunity cost rises

- As the opportunity cost of a good rises, the quantity demanded falls and the incentive to switch to a substitute becomes stronger

Substitution Effect & Income Effect

- ➢ Income effect
  - If price rises, your real income decreases
- When a price of a good rises, other things remaining the same, the price rises relative to income
- When people are faced with a higher price and an unchanged income, they decrease the quantity demanded of a good with the increase in price

Example: Substitution Effect & Income Effect

Price of an ice cream increases from \$3 to \$6 and suppose a close substitute of an ice cream is frozen yogurt

- People will tend to substitute frozen yogurt for ice cream, increasing the consumption of frozen yogurt and decreasing the consumption of ice cream ⟶ Substitution Effect
- Faced with tighter budget, people buy even fewer ice creams ⟶ Income Effect

The quantity of ice creams demanded decreases for these two reasons

Demand Schedule vs Demand Curve

- ➢ Demand schedule that shows quantity demanded at each price
- ➢ Demand curve is a graph of the relationship between the price of a good and the quantity demanded, ceteris paribus
    - Illustrates how the quantity demanded of the good changes as its price varies

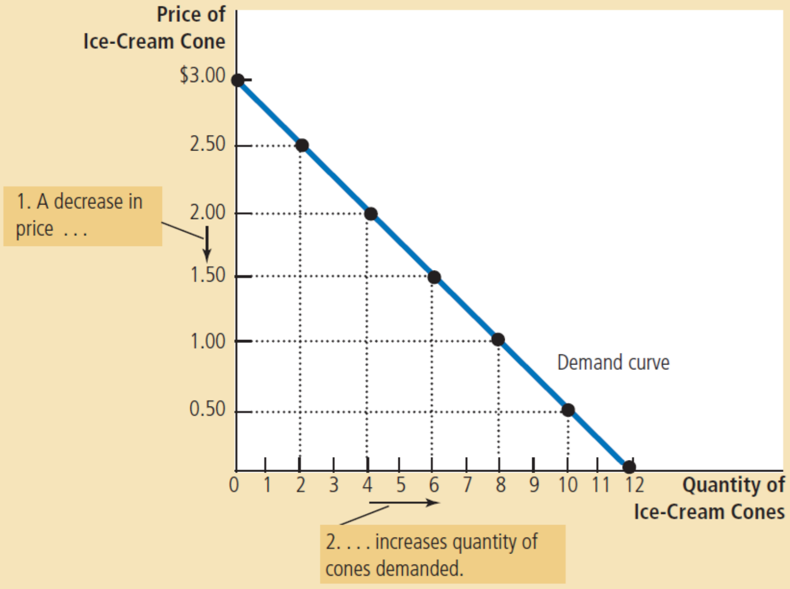
Demand Schedule vs Demand Curve

Figure 1

Catherine's Demand Schedule and Demand Curve

The demand schedule is a table that shows the quantity demanded at each price. The demand curve, which graphs the demand schedule, illustrates how the quantity demanded of the good changes as its price varies. Because a lower price increases the quantity demanded, the demand curve slopes downward.

Price of Ice-Cream Cone	Quantity of Cones Demanded
$0.00	12 cones
0.50	10
1.00	8
1.50	6
2.00	4
2.50	2
3.00	0



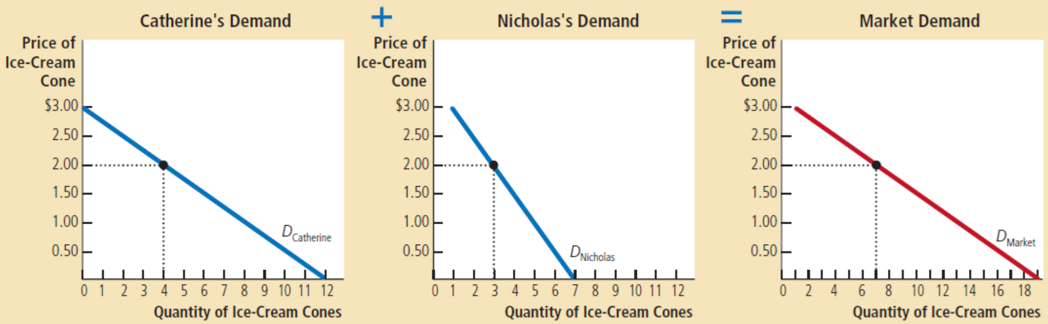
Market Demand vs Individual Demand

The quantity demanded in a market is the sum of the quantities demanded by all the buyers at each price. Thus, the market demand curve is found by adding horizontally the individual demand curves. At a price of $2.00, Catherine demands 4 ice-cream cones, and Nicholas demands 3 ice-cream cones. The quantity demanded in the market at this price is 7 cones.

Figure 2

Market Demand as the Sum of Individual Demands

Price of Ice-Cream Cone	Catherine		Nicholas		Market
$0.00	12	+	7	=	19 cones
0.50	10		6		16
1.00	8		5		13
1.50	6		4		10
2.00	4		3		7
2.50	2		2		4
3.00	0		1		1



Change in Quantity Demanded vs Change in Demand

- If the price of the good changes but no other factor influences the buying plan of a consumer, we illustrate the effect as a movement along the demand curve

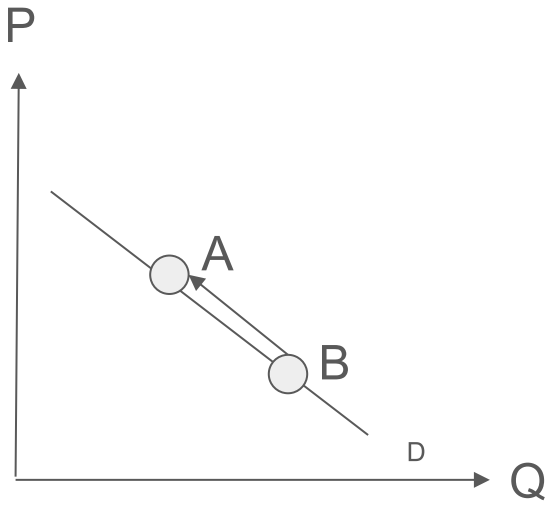
  Change in Quantity Demanded $\longrightarrow$ Movement along the Demand Curve

- If the price of a good remains constant but, some other factor influences the demand or buying plan of that good, then there will be a change in the demand causing the demand curve to shift

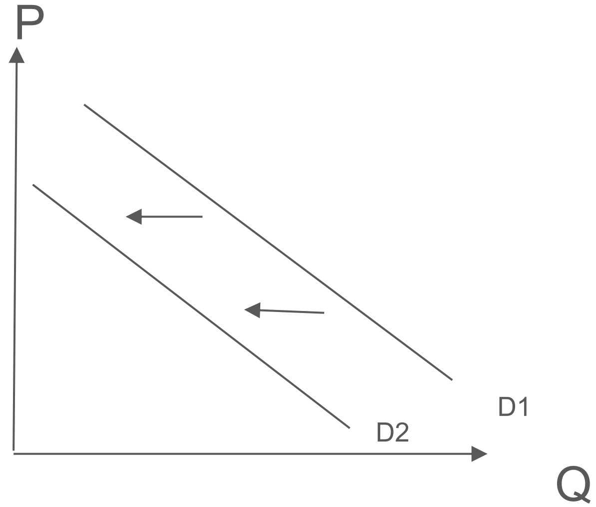
  Change in Demand $\longrightarrow$ Shift of the Demand Curve

Decrease in Quantity Demanded vs Decrease in Demand

Decrease in Quantity Demanded



Decrease in Demand



Shift in the Demand Curve

➢ When any factor that influences buying plans changes, other than the price of the good, there is a change in the demand

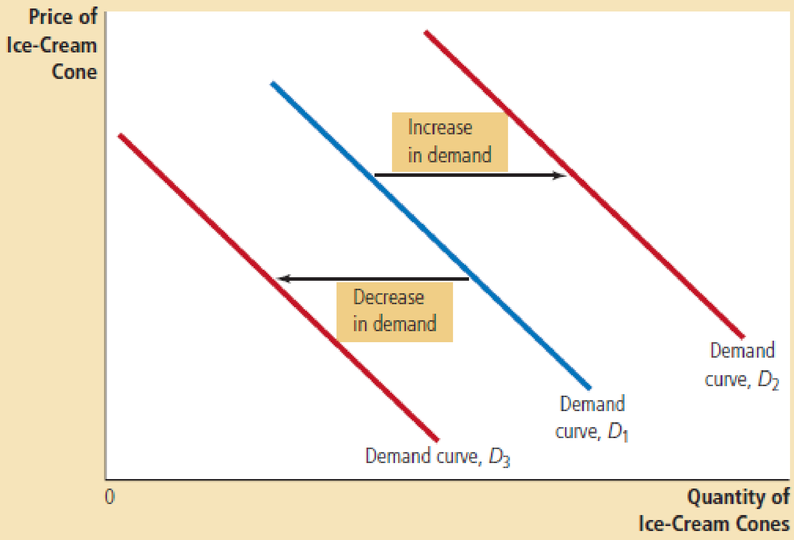
- When demand increases, the demand curve shifts to the right and the quantity demanded at each price is greater
- When demand decreases, the demand curve shifts to the left and the quantity demanded at each price is smaller

Shift in the Demand Curve

Figure 3

Shifts in the Demand Curve

Any change that raises the quantity that buyers wish to purchase at any given price shifts the demand curve to the right. Any change that lowers the quantity that buyers wish to purchase at any given price shifts the demand curve to the left.



Determinants of Demand

There are many factors that can cause change in demand. Six main factors are as follows:

- Preferences, tastes
- The price of related goods
- Expected future prices
- Income
- Expected future income and credit
- Populations

Determinants of Demand

➢ **Preferences, tastes**: Demand depends on preferences

➢ **The price of related goods:** Change in the price of one good influences the quantity demanded for the other good

- When two goods are **substitutes**, an increase in price of one good leads to an increase in the demand for the other good
  - A substitute is a good that can be used in place of another good
  - Example: ice cream & frozen yogurt, hot dogs & hamburgers, sweaters and sweatshirts

- When two goods are **complements**, an increase in price of one good leads to a decrease in the demand for the other good
  - A complement is a good that is used in conjunction with other good
  - Example: ice cream & hot fudge, coffee & creamer

Determinants of Demand

- **Expected future prices**: Your expectations about the future may affect your demand for a good or service today
  - If you expect to earn a higher income next month, you may choose to save less now and spend more of your current income buying ice cream
  - If you expect the price of ice cream to fall tomorrow, you may be less willing to buy an ice-cream cone at today's price

Determinants of Demand

- **Income**: Consumer's income influences demand. If the income of a consumer falls, he/she has less to spend on most of the good. However, a fall in income leads to a fall in the demand for most goods, but does not lead to a fall in the demand for all goods
  - A good for which, other things equal, a decrease (or an increase) in income leads to a decrease (or an increase) in demand is called a **normal good**
  - A good for which, other things equal, a decrease (or an increase) in income leads to an increase (or a decrease) in demand is called an **inferior good**

  Example: As income rises, the demand for air travel (a normal good) increases and the demand for long-distance bus trips (an inferior good) decreases

Determinants of Demand

- **Expected future income and credit**: When expected future income increases or credit becomes easier to get, demand for the good might increase now
  - Example: a salesperson gets the news that she will receive a big bonus at the end of the year, so she goes into debt and buys a new car right now, rather than wait until she receives the bonus
- **Population**: Demand also depends on the size and the age structure of the population
  - Example: a decrease in the college-age population would lead to a lower demand for college places, where as a rising aging population would mean a higher demand for nursing home services

Supply

Supply

- The **quantity supplied** of a good or service is the amount that producers plan to sell during a given time period at a particular price
- If a firm supplies a good or service, the firm
  - Has the resources and technology to produce it,
  - Can profit from producing it, and
  - Plans to provide it and sell it

Law of Supply

- ➢ The Law of Supply states that, other things remaining the same,
  - The higher the price of a good, the greater is the quantity supplied
  - The lower the price of a good, the smaller is the quantity supplied
- ➢ When the price of ice cream is high, selling ice cream is profitable, and so the quantity supplied is large
  - Sellers of ice cream work long hours, buy many ice-cream machines, and hire many workers
- ➢ By contrast, when the price of ice cream is low, the business is less profitable, so sellers produce less ice cream
  - At a low price, some sellers may even choose to shut down, and their quantity supplied falls to zero

Supply Schedule vs Supply Curve

- ➢ Supply schedule is a table that shows the relationship between the price of a good and the quantity supplied
- ➢ Supply curve is a graph of the relationship between the price of a good and the quantity supplied, holding everything else constant that influences how much producers of the good want to sell
    - It shows the relationship between the quantity supplied of a good and its price, when all the other influences on producer's planned sales remain the same

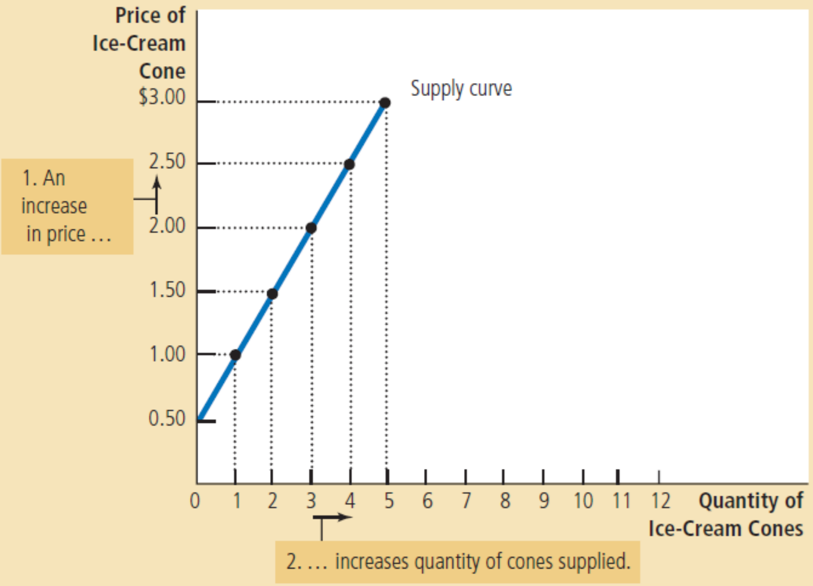
Supply Schedule vs Supply Curve

Figure 5

Ben's Supply Schedule and Supply Curve

The supply schedule is a table that shows the quantity supplied at each price. This supply curve, which graphs the supply schedule, illustrates how the quantity supplied of the good changes as its price varies. Because a higher price increases the quantity supplied, the supply curve slopes upward.

Price of Ice-Cream Cone	Quantity of Cones Supplied
$0.00	0 cones
0.50	0
1.00	1
1.50	2
2.00	3
2.50	4
3.00	5



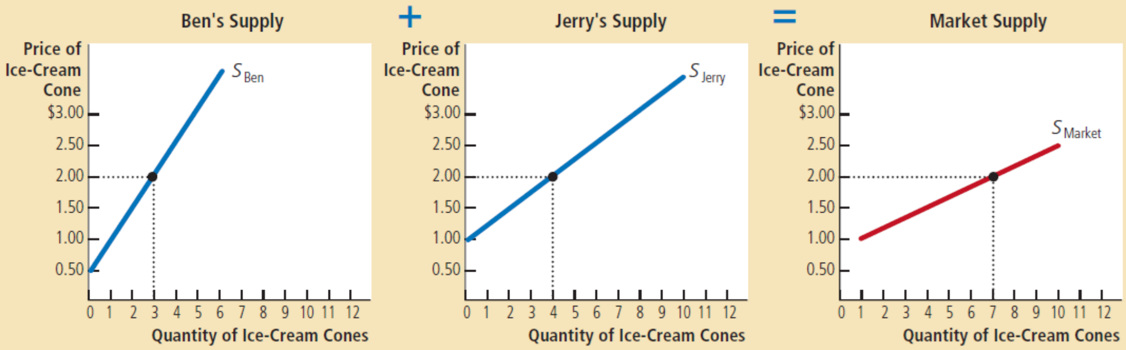
Market Supply vs Individual Supply

The quantity supplied in a market is the sum of the quantities supplied by all the sellers at each price. Thus, the market supply curve is found by adding horizontally the individual supply curves. At a price of $2.00, Ben supplies 3 ice-cream cones, and Jerry supplies 4 ice-cream cones. The quantity supplied in the market at this price is 7 cones.

Figure 6

Market Supply as the Sum of Individual Supplies

Price of Ice-Cream Cone	Ben		Jerry		Market
$0.00	0	+	0	=	0 cones
0.50	0		0		0
1.00	1		0		1
1.50	2		2		4
2.00	3		4		7
2.50	4		6		10
3.00	5		8		13



Change in the Quantity Supplied vs Change in Supply

- A point on the supply curve shows the quantity supplied at a given price, so a movement along the supply curve shows a change in the quantity supplied
- If the price of the good changes and other things remain the same, there is a change in the quantity supplied of that good

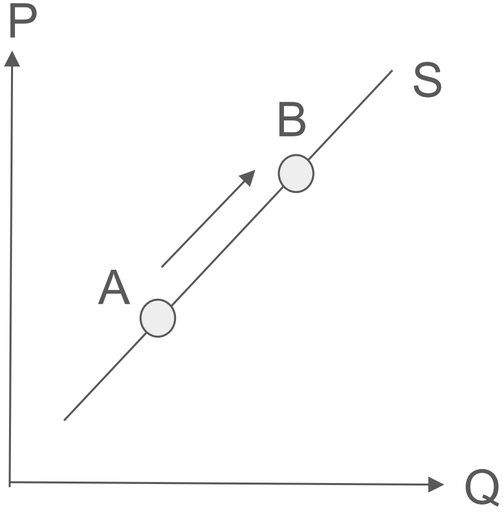
  Change in the Quantity Supplied ⟶ Movement along the Supply Curve

- A change in any factors that influences the selling plans (other than the price of that good), leads to a change in supply causing the supply curve to shift

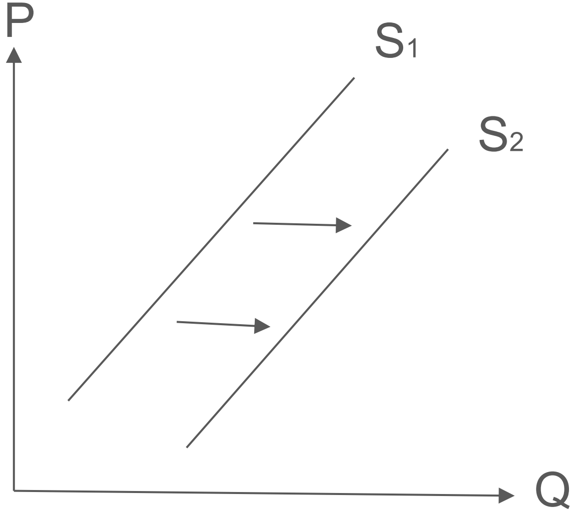
  Change in Supply ⟶ Shift of the Supply Curve

Change in the Quantity Supplied vs Change in Supply

Increase in Quantity Supplied



Increase in Supply



Shift in the Supply Curve

- When any factor that influences selling plans changes, other than the price of the good, there is a change in supply
    - When any change raises the quantity supplied at every price, the supply curve shifts to the right
    - When any change reduces the quantity supplied at every price, the supply curve shifts to the left

Shift in the Supply Curve

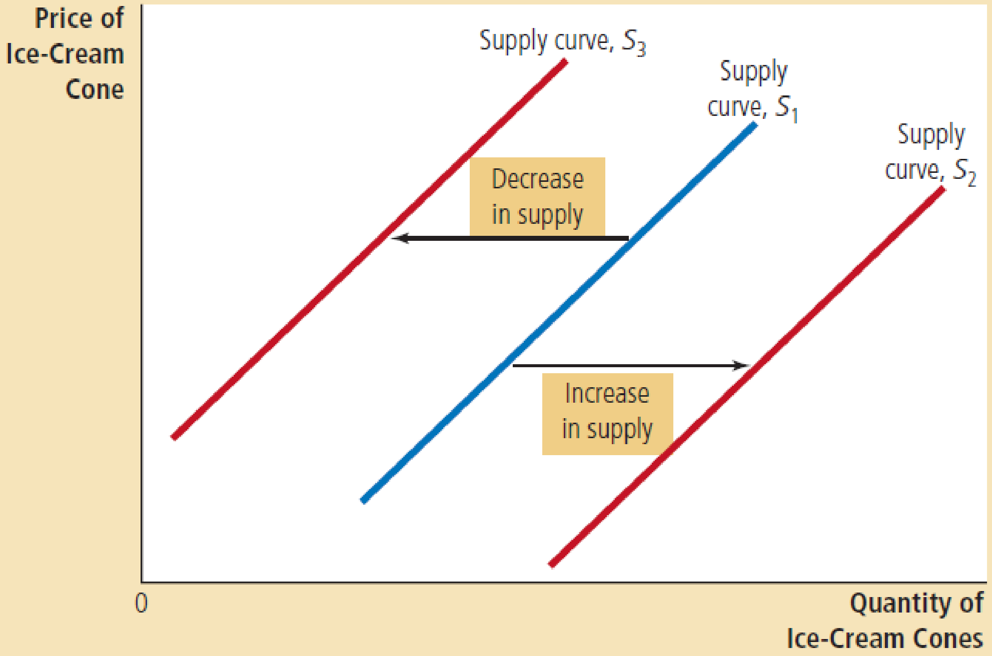


Figure 7

Shifts in the Supply Curve
Any change that raises the quantity that sellers wish to produce at any given price shifts the supply curve to the right. Any change that lowers the quantity that sellers wish to produce at any given price shifts the supply curve to the left.

Determinants of Supply

Some of the main factors that bring changes in supply are as follows:

- The prices of factors of production
- The prices of related goods produced
- Expected future prices
- The number of suppliers
- Technology
- The state of nature

Determinants of Supply

- **The prices of factors of production**: The prices of the factors of production used to produce a good influence its supply
  - Example: to produce ice cream various inputs are used, such as cream, sugar, flavouring, ice cream machines, the buildings in which the ice cream is made, and labour of workers to mix the ingredients and operate the machines
  - When the price of one or more of these inputs rises, producing ice cream becomes less profitable, and firms supply less ice cream
  - If input prices rise substantially, a firm might shut down and supply no ice cream at all

Determinants of Supply

➢ **The prices of related goods produced**: The prices of related goods that firms produce influence supply

- Example 1: if two goods such as energy bars and energy gels are substitutes in production (goods that can be produced by using the same resources), then if the price of energy gel rises, firms switch from production of energy bars to energy gels, increasing the supply of energy gels and decreasing the supply of energy bars

- Example 2: if two goods such as beef and cowhide are complements in production (goods that must be produced together), then if the price of beef rises, the supply of cowhide increases

Determinants of Supply

- **Expected future prices**: If the expected future price of a good rises, the return from selling the good in the future increases and is higher than it is today. So supply decreases today and increases in the future
  - For example, if a firm expects the price of ice cream to rise in the future, it will put some of its current production into storage and supply less to the market today
- **The number of suppliers**: The larger the number of suppliers/firms that produce a good, the greater is the supply of the good in that industry
  - As new firms enter an industry, the supply in that increases
  - As firms leave an industry, the supply in that industry decreases

Determinants of Supply

- **Technology**: A technology change occurs when a new method is discovered that lowers the cost of producing a good
  - Example: The invention of the mechanized ice-cream machine, reduced the amount of labor necessary to make ice cream. By reducing firms’ costs, the advance in technology raised the supply of ice cream
- **The state of nature**: The state of nature includes all the natural forces that influence production. It includes the state of the weather and, more broadly the natural environment
  - Good weather can increase the supply of many agricultural products and bad weather can decrease their supply

Market Equilibrium

Market Equilibrium

- We have seen that when the price of a good rises, the quantity demanded decreases and the quantity supplied increases
- We are now going to see how the price adjusts to coordinate buying and selling plans and achieve an equilibrium in the market
- **Equilibrium** is a situation in which the market price has reached the level at which quantity supplied equals quantity demanded
- **Equilibrium price** is the price that balances quantity supplied and quantity demanded
- **Equilibrium quantity** is the quantity supplied and the quantity demanded at the equilibrium price

Market Equilibrium

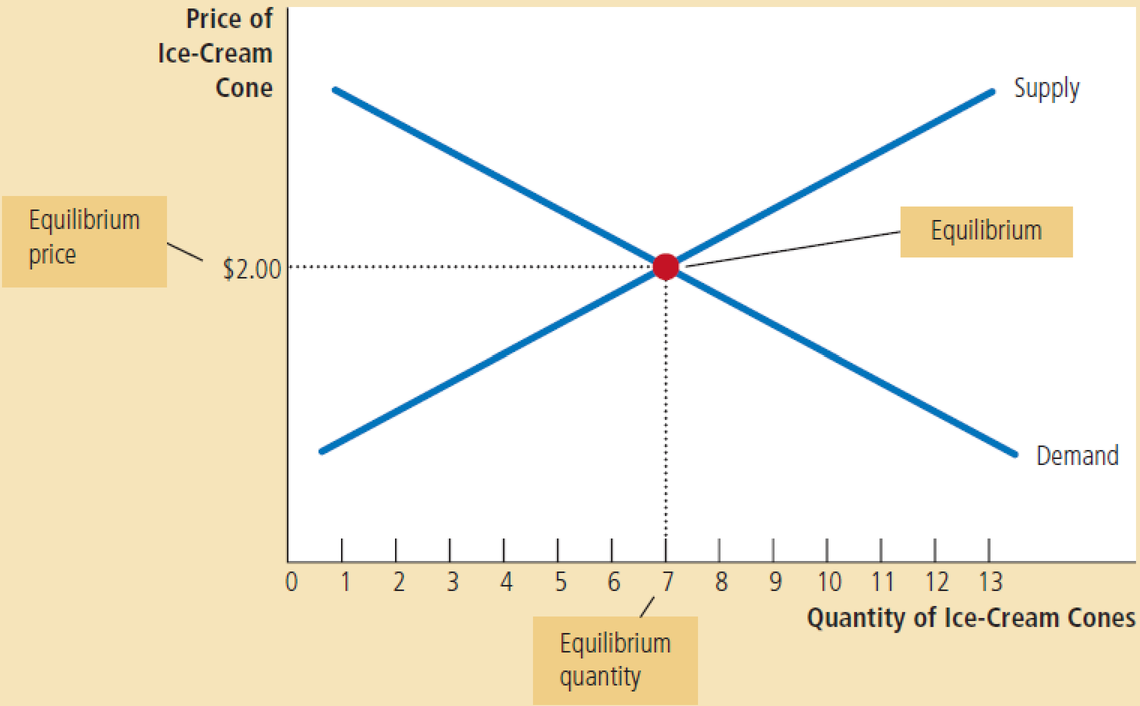


Figure 8

The Equilibrium of Supply and Demand
The equilibrium is found where the supply and demand curves intersect. At the equilibrium price, the quantity supplied equals the quantity demanded. Here the equilibrium price is $2.00: At this price, 7 ice-cream cones are supplied, and 7 ice-cream cones are demanded.

Price as a Regulator

- ➢ The price of a good regulates the quantities demanded and supplied
- ➢ If the price is too high, the quantity supplied exceeds the quantity demanded

  The quantity supplied exceeds the quantity demanded $\longrightarrow$ Surplus

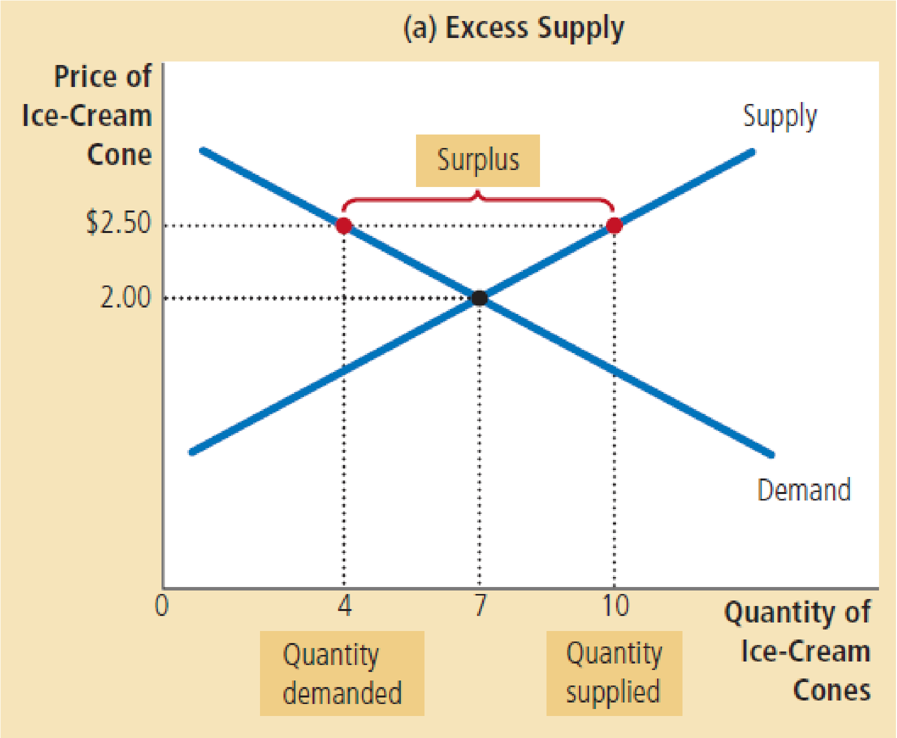
  - Surplus is a situation in which quantity supplied is greater than quantity demanded

- ➢ If the price is too low, the quantity demanded exceeds the quantity supplied

  The quantity demanded exceeds the quantity supplied $\longrightarrow$ Shortage

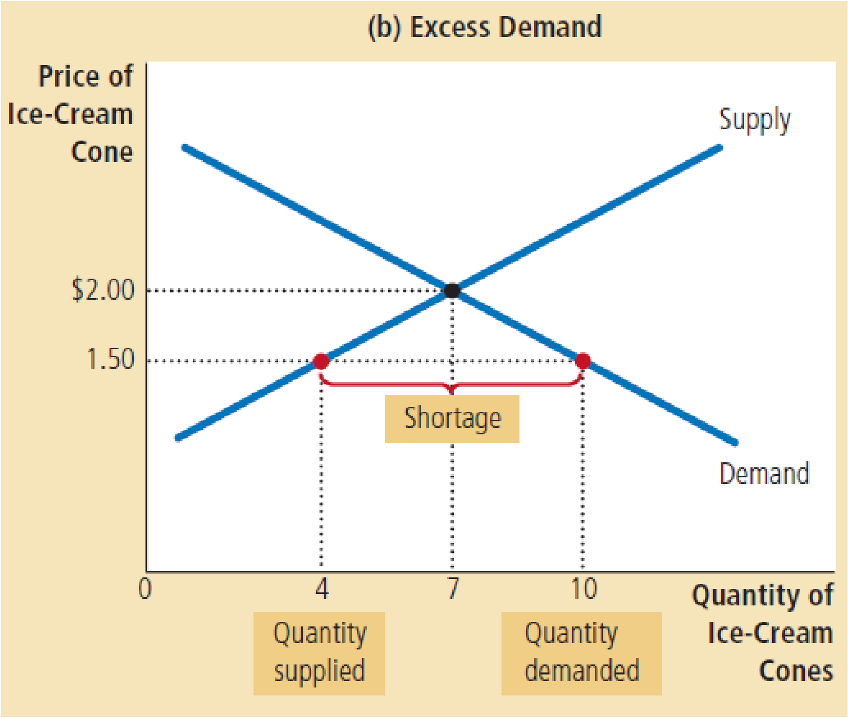
  - Shortage is a a situation in which quantity demanded is greater than quantity supplied

Excess Supply



- ➢ Suppose market price is $2.50 per cone (above equilibrium price) where the quantity of the ice cream supplied (10 cones) exceeds the quantity demanded (4 cones)
- ➢ Resulting in a surplus of ice cream
- ➢ Suppliers are unable to sell all they want at the going price
- ➢ So, they respond to the surplus by cutting their prices. Falling prices, in turn, increase the quantity demanded and decrease the quantity supplied
- ➢ These changes represent movements along the supply and demand curves
- ➢ Prices continue to fall until the market reaches the equilibrium

Excess Demand



- Suppose market price is $1.50 per cone (below the equilibrium price), where the quantity demanded of ice cream exceeds the quantity supplied
- Resulting in a shortage of ice cream
- With too many buyers chasing too few goods, sellers can respond to the shortage by raising their prices
- These price increases cause the quantity demanded to fall and the quantity supplied to rise
- Once again, these changes represent movements along the supply and demand curves, and they move the market toward the equilibrium

Market Equilibrium

- Thus, regardless of whether the price starts off too high or too low, the activities of the many buyers and sellers automatically push the market price toward the equilibrium price
- Once the market reaches its equilibrium, all buyers and sellers are satisfied, and there is no upward or downward pressure on the price
- How quickly equilibrium is reached varies from market to market depending on how quickly prices adjust
- In most free markets, surpluses and shortages are only temporary because prices eventually move toward their equilibrium levels

Example: Equilibrium, Surplus & Shortage

Price ($ per bar)	Quantity demanded (millions of bars per week)	Quantity supplied (millions of bars per week)	Shortage (-) or Surplus (+)
0.50	22	0	-22
1.00	15	6	-9
1.50	10	10	0
2.00	7	13	+6
2.50	5	15	+10

Changes in Equilibrium

- We have seen how supply and demand together determine a market's equilibrium, which in turn determines the price and quantity of the good that buyers purchase and sellers produce
- The equilibrium price and quantity depend on the position of the supply and demand curves
- When some event shifts one of these curves, the equilibrium in the market changes, resulting in a new price and a new quantity exchanged between buyers and sellers

Steps to Analyze Changes in Equilibrium

1. We figure whether the event shifts the supply curve, the demand curve, or, in some cases, both curves
2. We decide whether the curve shifts to the right or to the left
3. We use the supply-and-demand diagram to compare the initial and the new equilibrium, which shows how the shift affects the equilibrium price and quantity

Changes in Equilibrium - An Increase in Demand

Suppose due to very hot weather, demand for ice cream increases

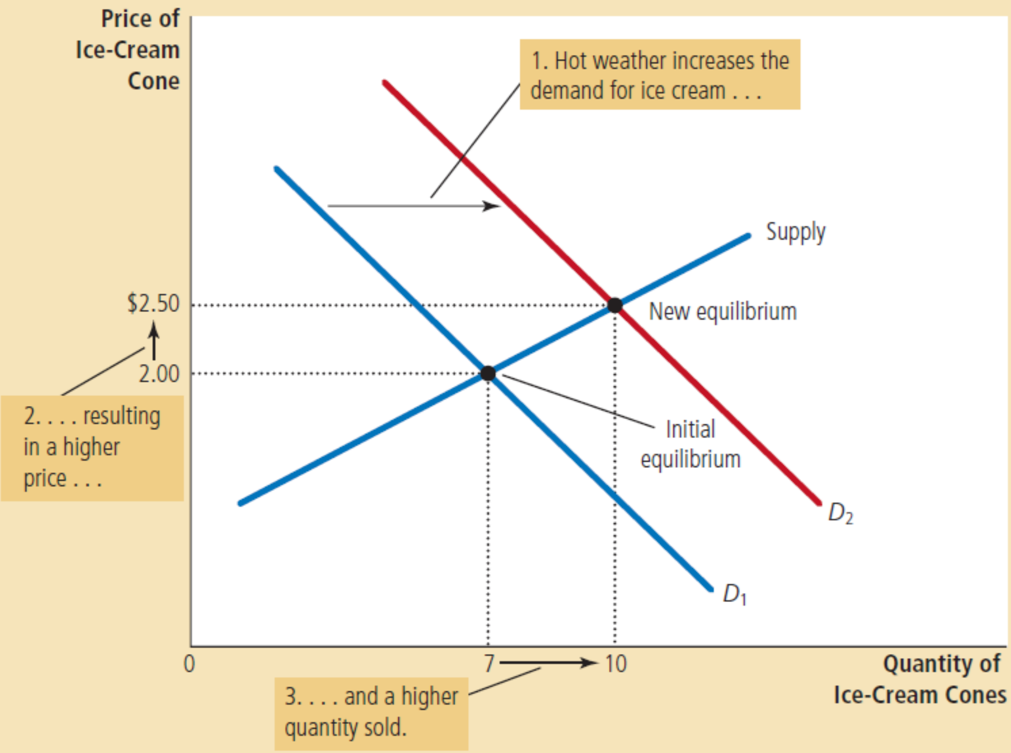
- ➢ Demand curve shifts to the right and supply curve is unchanged
- ➢ As a result, the quantity demanded of ice cream is higher at every price
- ➢ This leads to an excess demand for ice cream and this shortage induces firms to raise the price
- ➢ The increase in demand raises the equilibrium price and the equilibrium quantity
- ➢ In other words, the hot weather increases the price of ice cream and the quantity supplied of ice cream

Changes in Equilibrium - An Increase in Demand

Figure 10

How an Increase in Demand Affects the Equilibrium

An event that raises quantity demanded at any given price shifts the demand curve to the right. The equilibrium price and the equilibrium quantity both rise. Here an abnormally hot summer causes buyers to demand more ice cream. The demand curve shifts from D_1 to D_2, which causes the equilibrium price to rise from \$2.00 to \$2.50 and the equilibrium quantity to rise from 7 to 10 cones.



Changes in Equilibrium - A decrease in Supply

Suppose, a hurricane destroys part of the sugarcane crop and drives up the price of sugar. How does this event affect the market for ice cream?

- ➢ Increase in price of sugar (input for ice cream) raises the cost of production and in turn reduces the amount of ice cream that firms produce and sell at any given price
- ➢ As a result, supply curve shifts to the left (and demand curve is unchanged as higher cost of input does not directly affect the demand for ice cream)
- ➢ This shift in the supply curve will lead to an increase in equilibrium price and decrease in equilibrium quantity

Changes in Equilibrium - A decrease in Supply

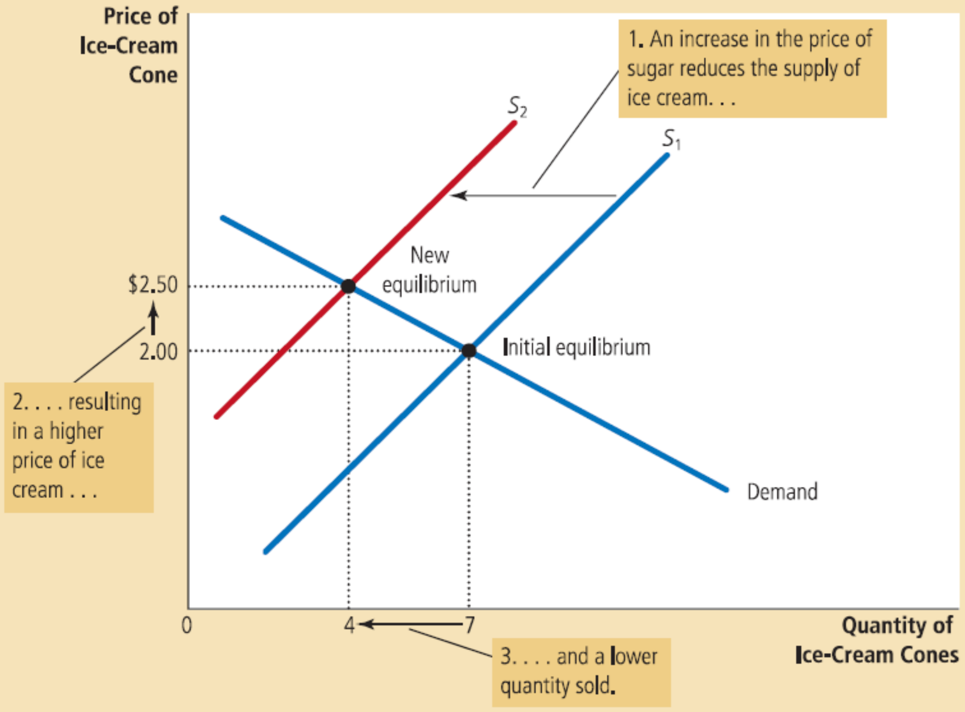


Figure 11

How a Decrease in Supply Affects the Equilibrium

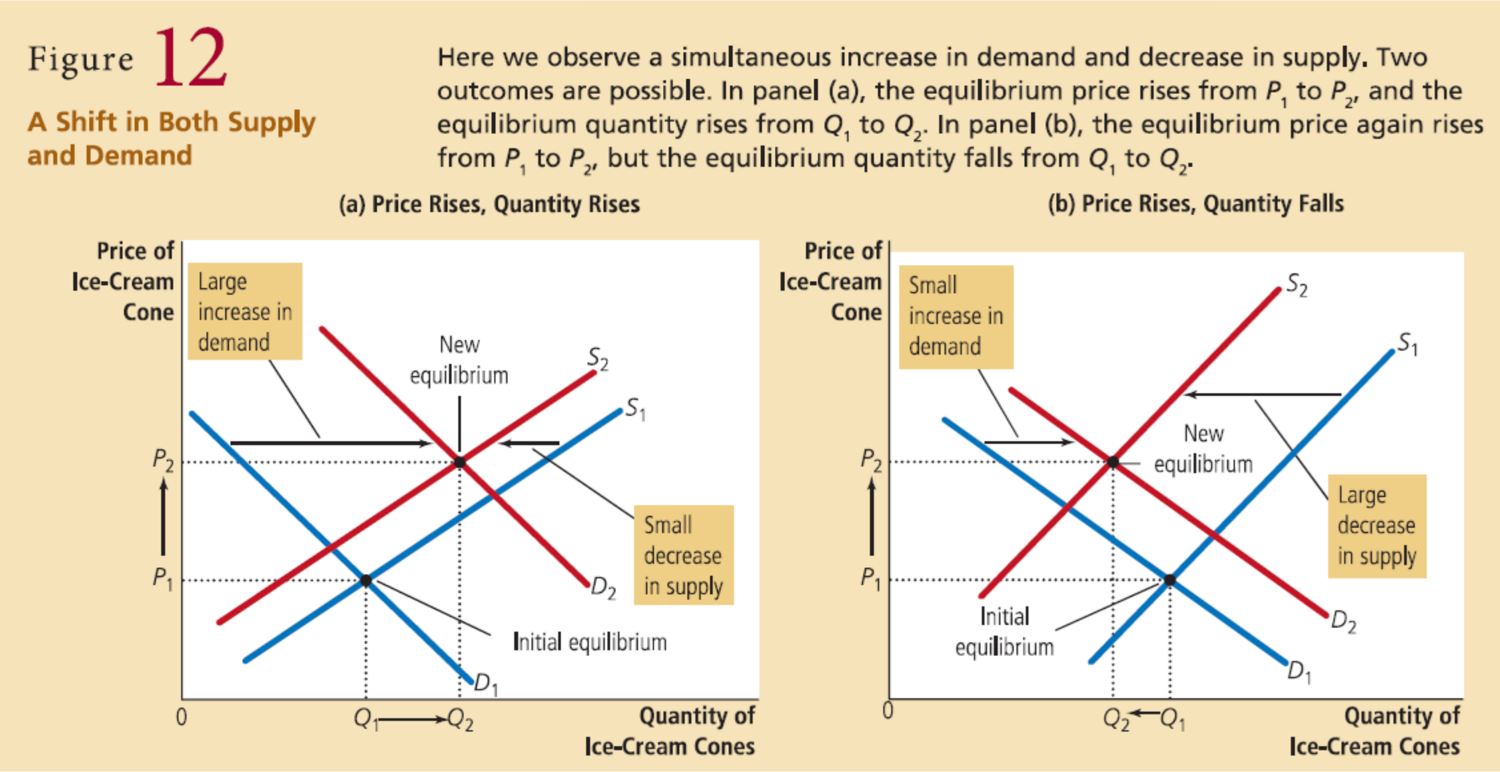
An event that reduces quantity supplied at any given price shifts the supply curve to the left. The equilibrium price rises, and the equilibrium quantity falls. Here an increase in the price of sugar (an input) causes sellers to supply less ice cream. The supply curve shifts from S_1 to S_2, which causes the equilibrium price of ice cream to rise from $2.00 to $2.50 and the equilibrium quantity to fall from 7 to 4 cones.

Changes in Equilibrium - Shifts in Both Demand & Supply

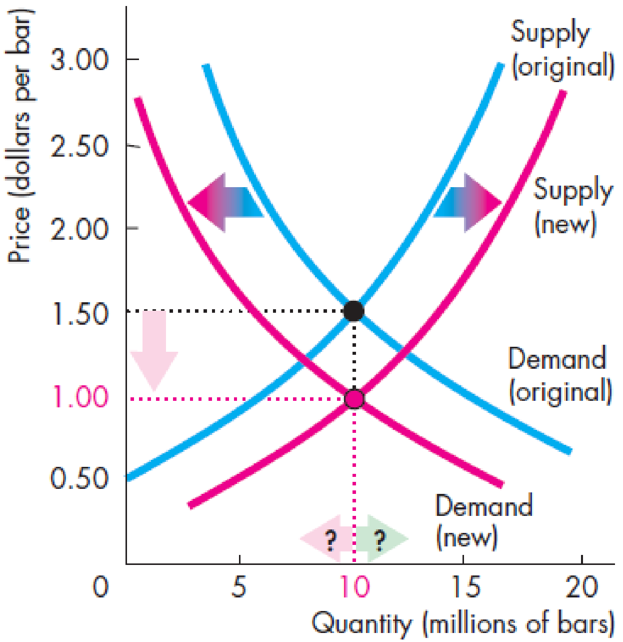
Suppose that a heat wave and a hurricane occur during the same summer

- ➢ Due to hot weather, the demand for ice cream rises causing the demand curve to shift to the right
- ➢ At the same time, due to hurricane sugar price rises, increasing the cost of producing ice cream and in turn causing the supply curve to shift to the left
- ➢ Now this simultaneous shifts in demand and supply curves may lead to two outcomes
    - I. Demand increases substantially while supply falls just a little (equilibrium price rises and equilibrium quantity rises)
    - II. Supply falls substantially while demand rises just a little (equilibrium price rises and equilibrium quantity falls)

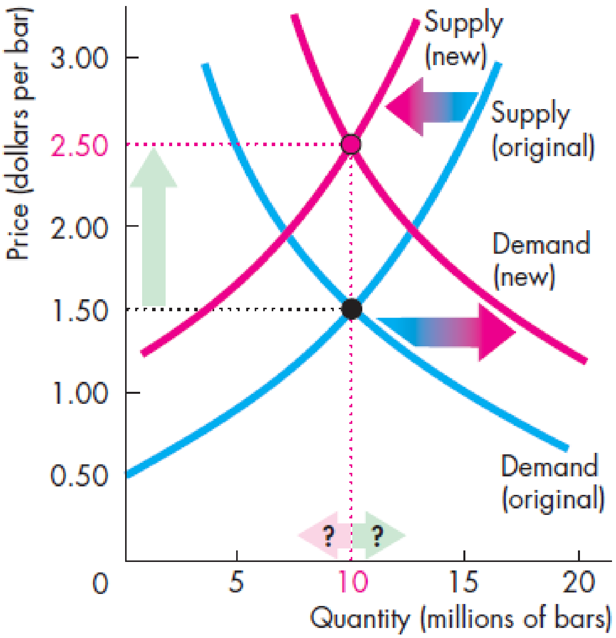
Changes in Equilibrium - Shifts in Both Demand & Supply



Changes in Equilibrium - Shifts in Both Demand & Supply

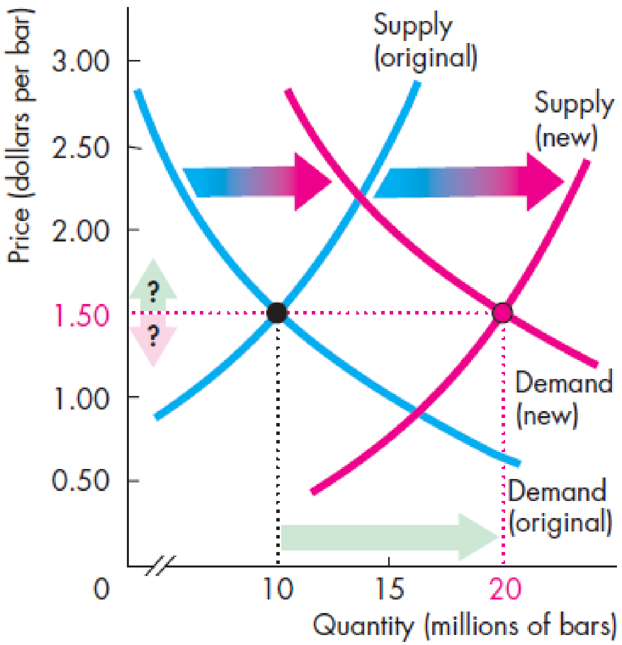


(f) Decrease in demand; increase in supply

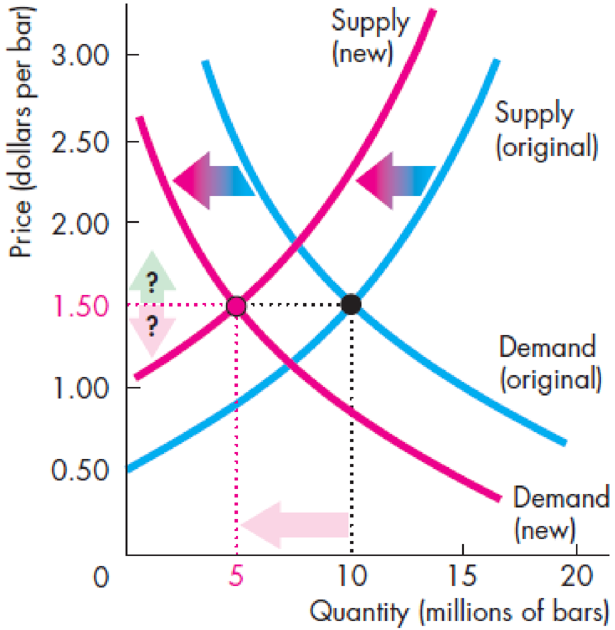


(h) Increase in demand; decrease in supply

Changes in Equilibrium - Shifts in Both Demand & Supply



(e) Increase in both demand and supply



(i) Decrease in both demand and supply

Exercise

Suppose the demand for kebabs in BRAC University at lunchtime is given by

$$Q_D = 100 - 10P$$

Kebab servers have the following supply plans

$$Q_S = 20P - 50$$

What is the equilibrium price and quantity?

Exercise - Solution

In equilibrium,

$$Q_D = Q_S$$

$$100 – 10P = 20P – 50$$

$$30P = 150$$

$$P_e = 5$$

Now we can solve for Q_e

$Q_e = 100 – 10P = 100 - 10 (5) = 50$

Exercise - Cont.

- ➢ What would you expect to happen in the market when the price is capped at Tk 7?
- ➢ What would you expect to happen in the market when the price is capped at Tk 3 instead?

*Workings demonstrated and explained in class